CRISPR-CAS9 GENE EDITING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on CRISPR-Cas9 Gene Editing

Question 1

10.0 points

Examine the diagram below that illustrates the Cas9 protein–sgRNA complex docking onto target DNA. Describe the molecular interactions between the sgRNA and the target DNA that are critical for specificity. Identify any key residues or structures in the Cas9 protein involved.

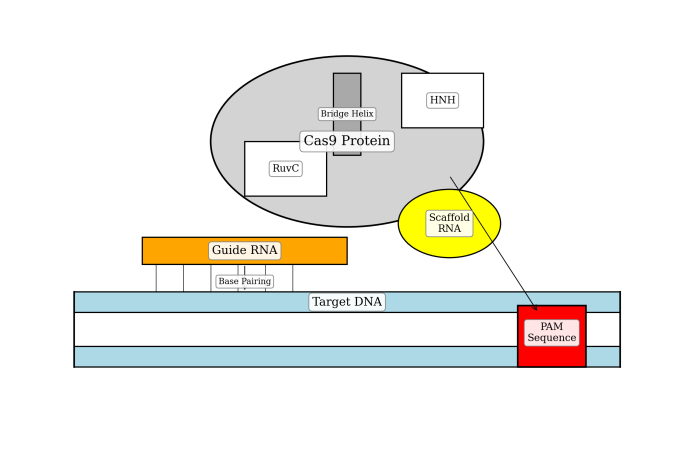

Fig. 1. Diagram of the Cas9 protein–sgRNA complex docking onto target DNA. Show the sgRNA aligned with the target DNA, highlighting the molecular interactions that ensure specificity. Include key residues or structures in the Cas9 protein involved in the interaction. Label the sgRNA, target DNA, and any critical residues or structures in the Cas9 protein that contribute to specificity.

Question 2

10.0 points

Analyze the diagram below illustrating the PAM sequence recognition and DNA unwinding mechanism in CRISPR-Cas9. Discuss the role of the PAM sequence in Cas9 activation and how DNA unwinding is achieved.

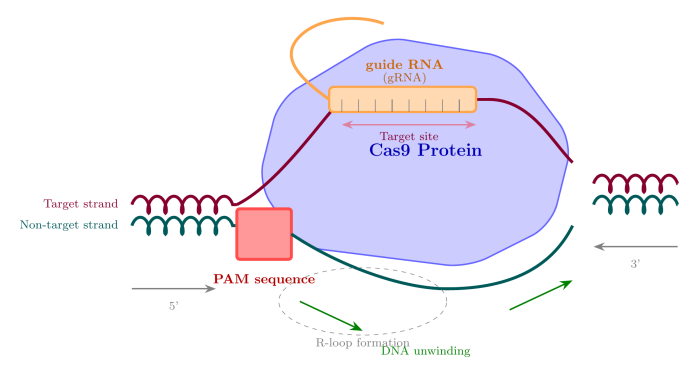

Fig. 2. Diagram illustrating the CRISPR-Cas9 mechanism, focusing on PAM sequence recognition and DNA unwinding. The diagram should show the Cas9 protein bound to a double-stranded DNA molecule. Highlight the PAM sequence on the DNA, which is essential for Cas9 activation. Include the guide RNA (gRNA) bound to the Cas9 protein, showing its complementary base pairing with the target DNA strand. Illustrate the unwinding of the DNA double helix at the target site, facilitated by the Cas9 protein, with arrows indicating the direction of unwinding. Label the PAM sequence, Cas9 protein, gRNA, and DNA strands clearly.

Question 3

10.0 points

Refer to the diagram below illustrating DNA double-strand break repair pathways: non-homologous end joining (NHEJ) versus homology-directed repair (HDR). Compare these pathways in terms of accuracy and potential applications in gene editing.

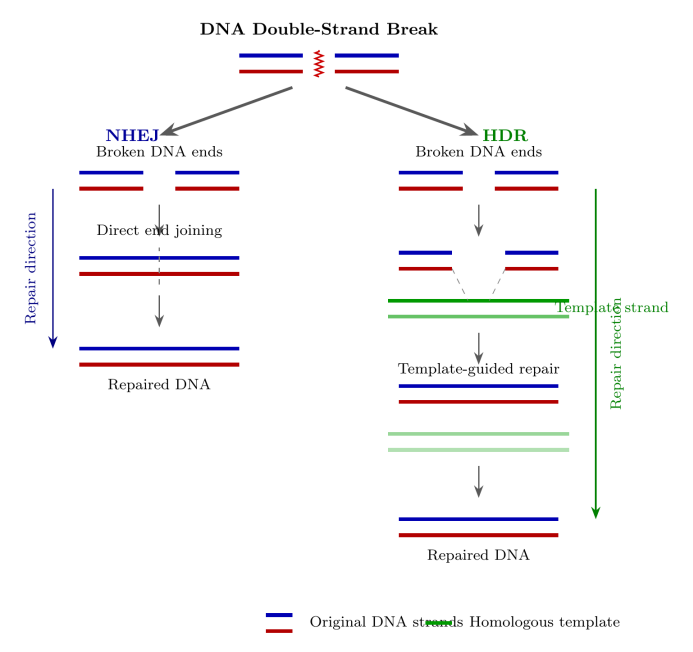

Fig. 3. Diagram illustrating DNA double-strand break repair pathways: non-homologous end joining (NHEJ) versus homology-directed repair (HDR). The diagram should show an initial DNA double-strand break at the top. Two arrows should branch from the break, one leading to the NHEJ pathway on the left and one to the HDR pathway on the right. For the NHEJ pathway: - Show the broken DNA ends. - Illustrate the binding of Ku proteins to the ends. - Show a step labeled 'End processing' where the ends are modified. - Show a final step labeled 'Ligation' resulting in 'Repaired DNA', with an annotation indicating it is error-prone or may contain insertions/deletions. For the HDR pathway: - Show the broken DNA ends. - Illustrate the binding of Rad51, which facilitates a 'Template search'. - Show a 'Homologous template' DNA strand being used. - Show a step labeled 'Template-guided synthesis' or 'Repair synthesis'. - Show the final 'Repaired DNA', with an annotation indicating it is an accurate, template-guided repair. Include clear labels for 'NHEJ', 'HDR', 'Ku proteins', 'Rad51', and 'Homologous template'. IMPORTANT: The diagram is missing required labels. Specifically: Missing label 'Ku proteins', Missing label 'End processing' (shown as '. The diagram is missing several key labels for proteins and process steps that were explicitly required by the description, such as 'Ku proteins', 'Rad51', 'End processing', and 'Ligation'.. Add each missing label as visible text at the appropriate location in the diagram.

Question 4

10.0 points

Analyze the mismatch tolerance in the diagram below for off-target effect prediction in CRISPR-Cas9 gene editing. How do different types of mismatches affect the likelihood of off-target effects?

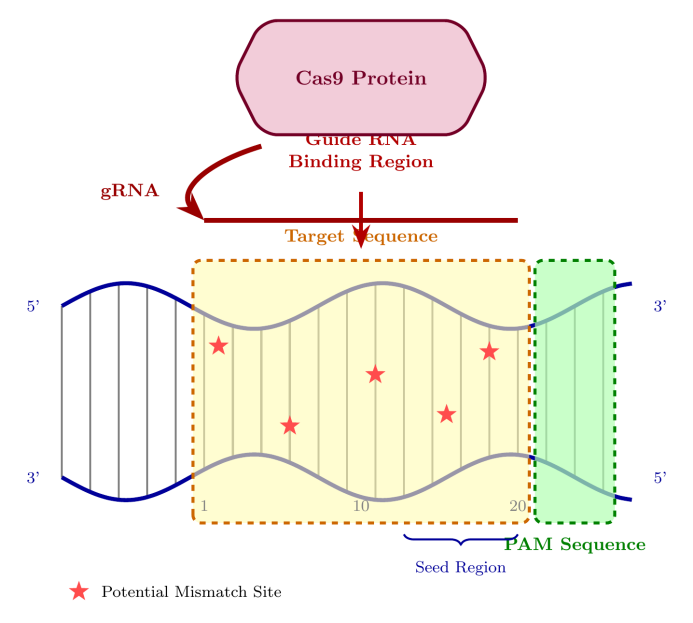

Fig. 4. Diagram of CRISPR-Cas9 complex interacting with target DNA sequence. Show the Cas9 protein, guide RNA (gRNA),

Question 5

10.0 points

Examine the base editing window in the diagram below for cytosine base editors (CBEs). Describe the CBE's base editing window and discuss the implications for specificity and efficiency in gene editing projects.

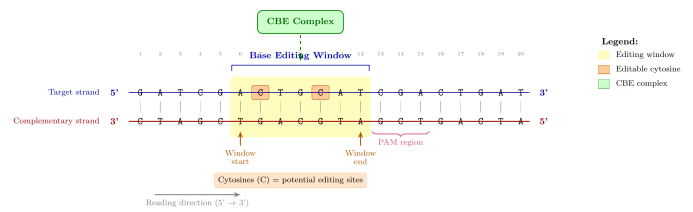

Fig. 5. A diagram showing a DNA double helix with a highlighted base editing window for cytosine base editors (CBEs). The DNA sequence should be represented with base pairs, and the editing window should be marked to indicate the range where cytosine bases can be edited. Label the cytosine bases within the editing window and show the directionality of the DNA strand (5' to 3'). Include annotations for the editing window boundaries and any relevant enzymes or proteins involved in the editing process.